PROJECT POSTER 3 OF 4 · THE IDEA & COMPARISON

Ask your documents. Trust the answer.

Why we built Lexi AI, how it works, and how it compares with the alternatives.

THE PROBLEM

Contracts, SOPs and reports pile up faster than anyone can read them. Pasting them into a general chatbot means **sending sensitive files to a third party**, getting **answers you can't check**, and leaving **no record** of who asked what.

THE IDEA

Bring the model to the documents, not the documents to the model. Retrieve the right passages first, let a local LLM answer **only from them**, cite and verify every claim – and log every action.

A HOW IT WORKS

1 Documents
PDF · DOCX · XLSX · PPTX · CSV · EML · audio (Whisper)

2 Ingest & guard
Upload validation, prompt-injection scan, true page provenance

3 Pre-process & chunk
NLTK clean-up, 3-sentence overlapping chunks

4 Index
MiniLM embeddings → FAISS vectors, plus a BM25 keyword index

5 Hybrid retrieval
Blend both scores → top-5 passages with page and relevance

6 Local LLM
Llama 3.2 3B via Ollama answers only from the passages; refuses below a threshold

OUTPUTS: Cited Q&A · Executive briefings · Compliance checks · Entities & clauses · Action items · ANN sentence tagging

Platform foundation: Flask web app + JSON API · PostgreSQL / SQLite · Alembic migrations · role-based access · API keys · hash-chained audit log · Stripe billing · 163 automated tests, CI on every push

B HOW IT COMPARES

WHAT MATTERS	READING IT YOURSELF	A GENERAL-PURPOSE CLOUD CHATBOT	LEXI AI
Searches a whole document set	◐ Slow and tiring	◐ Limited by paste size	✓ Hybrid search over every page
Cites document + page you can check	✓ By hand	✗ Rarely verifiable	✓ Numbered sources, page, relevance
Says "not found" instead of guessing	✓ Human judgement	◐ Often answers anyway	✓ Similarity threshold 95% balanced accuracy at a tuned threshold (Poster 1)
Documents stay on your machine	✓ Yes	✗ Sent to a provider	✓ Local LLM, optional zero-retention
Tamper-evident record of who did what	✗ Manual notes	✗ Not by default	✓ Hash-chained audit log
Repeatable compliance checklist	◐ Checklists by hand	◐ Ad-hoc prompts	✓ PASS / REVIEW / FAIL + verified quotes
Team roles, API and billing	✗ –	◐ Account-level only	✓ Roles, API keys, seat-based plans

✓ yes ◐ partly ✗ no · A qualitative comparison of approaches in general, not a benchmark of any named product.

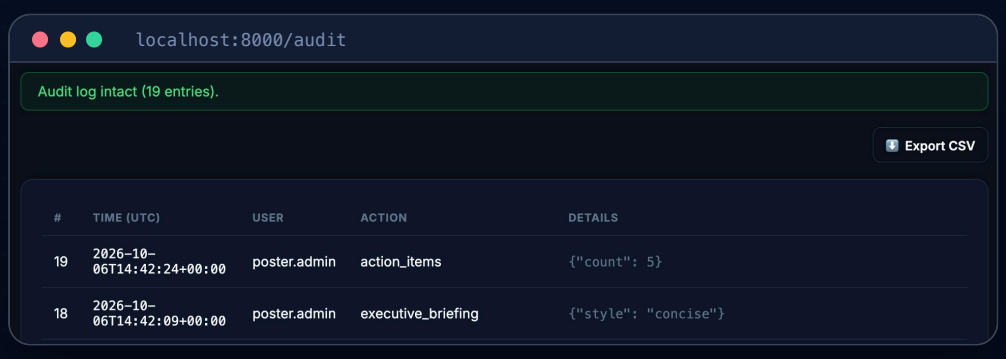
Private by design
Local model, optional zero-retention mode, and an audit log that stores metadata only – never document text.

Verifiable by design
Inline citations, shown confidence, verbatim-quote checks – and a human review lane when the model can't prove it.

Enterprise-ready
Roles, API keys, database migrations, seat-based billing and a CI-tested codebase.

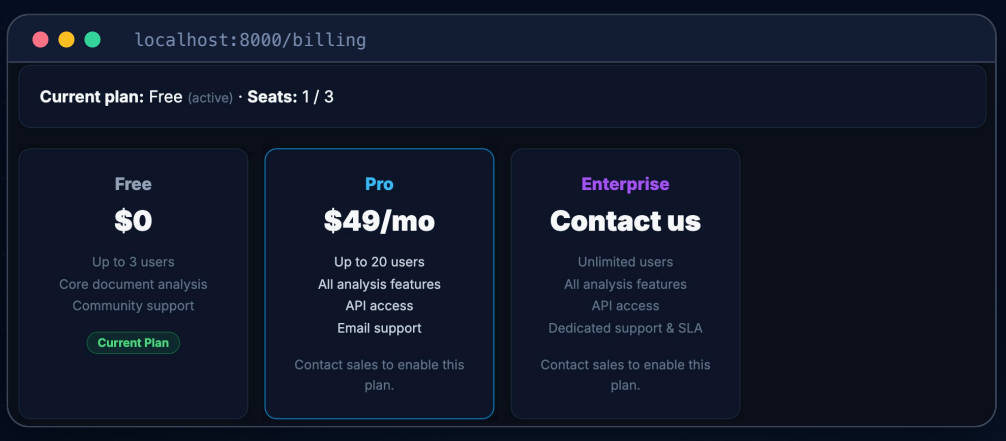
C Audit log – "who did what", tamper-evident

A hash chain flags any edit to history. Real log from our demo run: 19 entries, chain intact; details are metadata only.



D Plans & seat limits – built for teams

Free, Pro and Enterprise tiers with seat limits. Stripe checkout switches on when keys are configured; shown here in its unconfigured state.



SHIPPED
Cited Q&A · hybrid search · briefings · compliance · entities · action items · ANN tagging · roles & audit log · billing

NEXT
Single sign-on (SSO) · multi-tenancy for many organisations

LATER
Hosted deployment · monitoring · optional hosted-LLM tier